❄️ WINTER ARC: ML ENGINEERING ROADMAP (2025)

Student: Naif Noor

Duration: Dec 9, 2024 – Jan 7, 2025

Goal: Industry-Ready ML Engineer (Full Stack: Logic, ML, Deployment)

🛑 THE ANTI-LOST PROTOCOLS

1. **One Folder Per Day:** You must create a new folder (e.g., `Day_01_Logistic_Regression`) every morning. All code goes there.
2. **No AI for Logic:** You can use AI to explain error messages. You **cannot** use AI to write `for` loops, SQL queries, or Class definitions. You must type them.
3. **The "Done" Definition:** You do not sleep until the ✅ **Deliverable** is on your screen.
4. **The 45-Minute Rule:** If stuck on a bug for 45 mins, ask AI for the *syntax fix* only, then type it yourself.

🗓️ WEEK 1: THE FOUNDATION & UI

Focus: Classification, Metrics, Streamlit, and Python Basics.

Dec 09 (Mon): Logistic Regression

- **ML Task:** Watch "Logistic Regression Geometric Intuition" (CampusX). Implement it on "Social Network Ads" dataset.
- 🧠 **Logic Gym:** Create a list `[1, 2, 3, 4, 5]`. Write a `for` loop to square each number without using `numpy` or `map`.
- ✅ **Deliverable:** A screenshot of the `accuracy_score` printed in your terminal.

Dec 10 (Tue): Metrics (The Truth)

- **ML Task:** Watch "Confusion Matrix", "Precision/Recall". Plot a Heatmap using Seaborn.
- 🧠 **Logic Gym:** Create a Dictionary `{'A': 10, 'B': 20}`. Write code to swap keys and values manually
- →
- `{10: 'A', 20: 'B'}`.
- ✅ **Deliverable:** A saved image file `confusion_matrix.png`.

Dec 11 (Wed): KNN & EDA

- **ML Task:** Watch "KNN Intuition". Download **Heart Disease Dataset**. Check for missing values (`isnull().sum()`). Train KNN.
- 🧠 **Logic Gym:** Write a function `is_palindrome(word)` that returns `True/False` (String manipulation).
- ✅ **Deliverable:** A verifiable prediction: `model.predict([sample_data])` returns 0 or 1.

Dec 12 (Thu): Decision Trees

- **ML Task:** Watch "Decision Tree Entropy". Train `DecisionTreeClassifier`. Use `sklearn.tree.plot_tree` to visualize it.
- 🧠 **Logic Gym: OOP Basics.** Create a class `Dog` with an `__init__` method taking `name` and `age`, and a method `bark()`.
- ✅ **Deliverable:** A `.png` image showing the actual Tree flowchart.

Dec 13 (Fri): Streamlit Interface

- **ML Task:** Read Streamlit Docs. Create `app.py`. Add Title, 3 Input Fields, and a Button.
- 🧠 **Logic Gym: SQL Basics.** Go to LeetCode SQL 50. Solve "Recyclable and Low Fat Products" (Select/Where).
- ✅ **Deliverable:** Screenshot of your browser showing the app running on `localhost:8501`.

Dec 14 (Sat): Integration

- **ML Task:** Save model using `joblib`. Load it in `app.py`. Connect the button to the model.
- 🧠 **Logic Gym: SQL Aggregates.** Solve a problem using `COUNT`, `MAX`, or `AVG`.
- ✅ **Deliverable:** Video of you entering numbers in the website and getting a prediction result.

🗓️ WEEK 2: THE PROFESSIONAL UPGRADE

Focus: Ensembles, Pipelines, and API (Backend).

Dec 16 (Mon): Random Forest

- **ML Task:** Watch "Random Forest Intuition". Train RF and compare accuracy vs Decision Tree.
- 🧠 **Logic Gym: NumPy Basics.** Create a 5x5 matrix of zeros. Change the center 3x3 values to ones.
- ✅ **Deliverable:** Print Statement: `RF Accuracy: 0.85, DT Accuracy: 0.78`.

Dec 17 (Tue): Boosting Concepts

- **ML Task:** Watch "AdaBoost Intuition". Implement `AdaBoostClassifier`.
- 🧠 **Logic Gym: NumPy Math.** Perform Matrix Multiplication (Dot Product) of two arrays.
- ✅ **Deliverable:** Code runs without errors.

Dec 18 (Wed): XGBoost (The King)

- **ML Task:** Watch "XGBoost Math" (StatQuest). Train `XGBClassifier` with `GridSearchCV`.
- 🧠 **Logic Gym: Pandas.** Group data by a column (e.g., 'Sex') and find the mean of another (e.g., 'Age').
- ✅ **Deliverable:** Save the best model as `xgb_model.joblib`.

Dec 19 (Thu): Pipelines

- **ML Task:** Watch "Sklearn Pipelines". Refactor code: `Imputer` -> `Scaler` -> `Model`.
- 🧠 **Logic Gym: Python Lambda.** Write a lambda function to clean a string (e.g., remove spaces).
- ✅ **Deliverable:** `pipeline.predict(raw_data)` works without manual scaling.

Dec 20 (Fri): FastAPI (The Backend)

- **ML Task:** Watch "FastAPI for Beginners". Create `main.py` with a simple "Hello World" endpoint.
- 🧠 **Logic Gym: SQL Joins.** Solve a LeetCode problem requiring `LEFT JOIN`.
- ✅ **Deliverable:** Browser shows `{"Hello": "World"}` at `localhost:8000`.

Dec 21 (Sat): Microservice Connect

- **ML Task:** Update API to load `pipeline.joblib`. Create a POST endpoint `/predict`.
- 🧠 **Logic Gym: SQL Grouping.** Solve a problem using `GROUP BY` and `HAVING`.
- ✅ **Deliverable:** Screenshot of Swagger UI (`/docs`) showing a successful prediction.

📅 WEEK 3: THE STAND-OUT SKILLS

Focus: Unsupervised Learning, Docker, and GitHub.

Dec 23 (Mon): K-Means Clustering

- **ML Task:** Watch "K-Means". Cluster "Mall Customers Dataset".
- 🧠 **Logic Gym: OOP Methods.** Create a class `BankAccount` with `deposit()` and `withdraw()` logic.
- ✅ **Deliverable:** Scatter plot showing 5 different colored clusters.

Dec 24 (Tue): Market Basket Analysis

- **ML Task:** Learn "Apriori Algorithm". Use `mlxtend` to find rules (Bread -> Butter).
- 🧠 **Logic Gym: OOP Inheritance.** Create `SavingsAccount` that inherits from `BankAccount`.
- ✅ **Deliverable:** Print top 5 association rules.

Dec 25 (Wed): Docker Setup

- **ML Task:** Install Docker Desktop. Create a `Dockerfile` for your API.
- 🧠 **Logic Gym: String Manipulation.** Solve LeetCode "Valid Anagram" (Easy).
- ✅ **Deliverable:** Docker Engine status says "Running".

Dec 26 (Thu): Docker Build

- **ML Task:** Run `docker build -t my-api .` in terminal. Fix requirements errors.
- 🧠 **Logic Gym: Recursion.** Write a factorial function using recursion.
- ✅ **Deliverable:** Terminal says "Successfully built...".

Dec 27 (Fri): Docker Run

- **ML Task:** Run `docker run -p 8000:8000 my-api`. Test it in browser.
- 🧠 **Logic Gym: File I/O.** Write a script to read a CSV file using only the `csv` module (no Pandas).
- ✅ **Deliverable:** Your API works, but running inside a container.

Dec 28 (Sat): Git Push

- **ML Task:** Create GitHub Repo. Push all code. Add a professional `README.md`.
- 🧠 **Logic Gym: Debugging.** Pick a piece of old code, break it, and use the VS Code Debugger to fix it.
- ✅ **Deliverable:** A live GitHub link sent to yourself.

🗓️ WEEK 4: THE CAPSTONE PROJECT

Project: Telco Customer Churn Prediction (End-to-End).

- **Dec 30 (Mon):** Find Data (Kaggle). Clean Data. **Logic:** Pivot Tables in Pandas.
- **Dec 31 (Tue):** Build Pipeline (XGBoost + SMOTE). **Logic:** Python Sets (Operations).
- **Jan 01 (Wed):** Build FastAPI Backend. **Logic:** JSON parsing.
- **Jan 02 (Thu):** Build Streamlit Dashboard (calling the API). **Logic:** Review OOP.
- **Jan 03 (Fri):** Dockerize the application. **Logic:** Review SQL.
- **Jan 04 (Sat):** Deploy to **Render.com** (Cloud). **Deliverable:** A Live URL.
- **Jan 05 (Sun):** Documentation & Resume Update.
- **Jan 06 (Mon):** Final Polish. Interview Prep.
- **Jan 07 (Tue): SEMESTER STARTS.**